

JBM9204 INTRODUCTION TO BIOMETRIC SYSTEMS

UNIT-III

FACE RECOGNITION

Introduction to Face Recognition

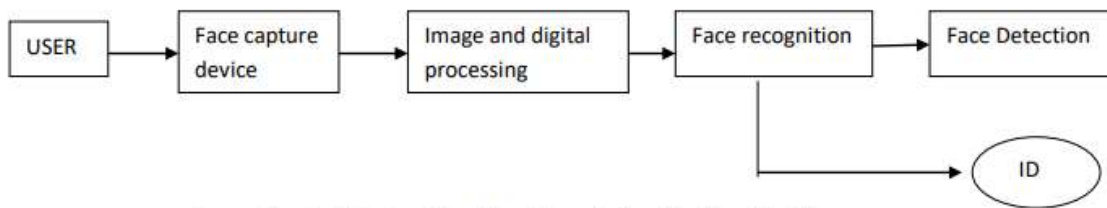
Among the various biometric identification methods, face recognition is one of the most flexible, working even when the subject is unaware of being scanned.

It also shows promise as a way to search through masses of people who spent only seconds in front of a "scanner" - that is, an ordinary digital camera.

Face recognition systems work by systematically analyzing specific features that are common to everyone's face - **the distance between the eyes, width of the nose, position of cheekbones, jaw line, chin and so forth.** These numerical quantities are then combined in a single code that uniquely identifies each person.

Biometric Authentication by face recognition

The following figure describes an access control system based on face authentication. In this model, each user has an account and a corresponding ID in the Face Database. On a user logging in the system, Face Authentication will use face recognition technologies to analyze and determine his ID as well as his permissions on the system.



Access Control System Based on Face Authentication Model

This model can be applied to access control systems where the number of people is small; for example, user accounts in an operating system, members of an office or a family.

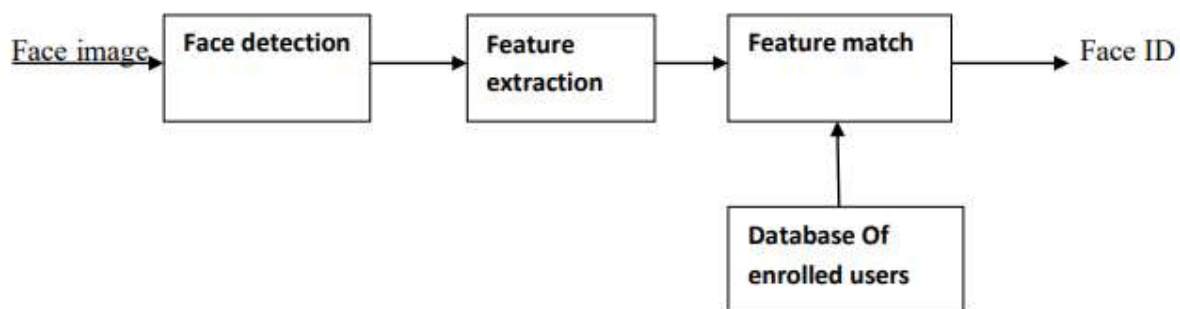
- When receiving a request, an access control system based on face authentication must find out exactly whether the person requesting is a client or an impostor.
- Right after that, it decides whether to accept the login or to treat him/her as an impostor and cry out "access denial"

- In order for Face Authentication to satisfy all the security issues that an access control system asks for, the face recognition algorithms in operation must be almost completely exact.

Face Recognition Model

As you can see from the diagram below, face recognition requires a wide range of technologies. Face recognition systems in general, and access control systems based on face authentication in particular, use a “learning” mechanism to collect data on facial characteristics of users.

Hence, the first important point to care about in a face recognition model is the Face Database storing this information.



When the system finishes scanning a video or photo of a user’s face, the digitalized information will go through these following modules one after another:

Face Detection

Face detection is the first step the engine takes to confirm the presence of faces as they appear on a live camera feed, a video recording or as it scans still image captures. The whole field of view is scanned for any area containing full or even partial human faces. Fast and precise face detection is a critical first step to ensure the performance of the entire facial recognition process.

Face Feature Extraction

Handled by another deep neural network model (referred to as the AI model), this step converts the detected face data into vector values in a high-dimensional space, known as facial feature values. Due to the nature of vectors, the similarity between two facial feature values can be assessed by calculating the distance between the two vectors. The task of the AI model is to group feature values belonging to the same

person as closely as possible in the spatial domain while separating those of different individuals.

During the training process, the AI model utilizes a vast amount of training data (over tens of millions), a large number of computation units (from hundreds of millions to billions), and extremely high-dimensional feature values (over 1024 dimensions). This allows the AI model to learn how to accurately analyze facial features and classify faces in a high-dimensional space. The training process often requires hundreds to thousands of GPU hours and involves iterative adjustments and optimizations of numerous model parameters and training settings, making it an extremely challenging task.

Comparison and Search

By calculating the similarity between two facial feature values and comparing that similarity to a predefined threshold, it can be determined whether the two faces belong to the same person. In the context of eKYC (electronic Know Your Customer), identity verification refers to comparing the face on an ID with the face in front of the camera to complete the verification process, which is an application of 1:1 facial match comparison.

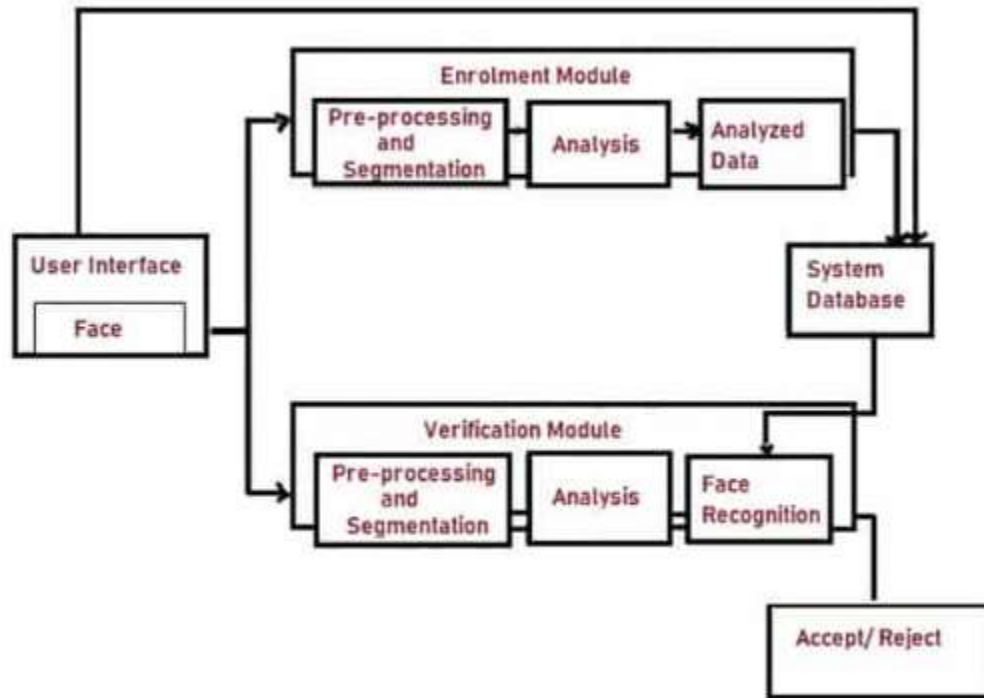
Additionally, if the facial feature values of faces appearing in a frame are compared against multiple facial feature values stored in a database, this constitutes 1: N facial recognition. Common applications include access control and attendance systems.

Difference between face detection and recognition

Aspect	Face Detection	Face Recognition
Definition	Process of finding and locating faces in an image or video.	Process of identifying or verifying <i>whose</i> face it is.
Output	Gives the position of face(s) with bounding boxes.	Gives the identity/label of the person or verifies match (Yes/No).
Complexity	Simpler – just finds presence of faces.	More complex – requires feature extraction and comparison with database.
Example	Camera app highlights a face with a rectangle.	Phone unlocks when it recognizes the owner’s face.

Components of Facial Recognition Systems

The diagram illustrates the working principle of a face recognition system, which is widely used for identity verification and authentication. The system works in two main phases: **enrolment** and **verification**.

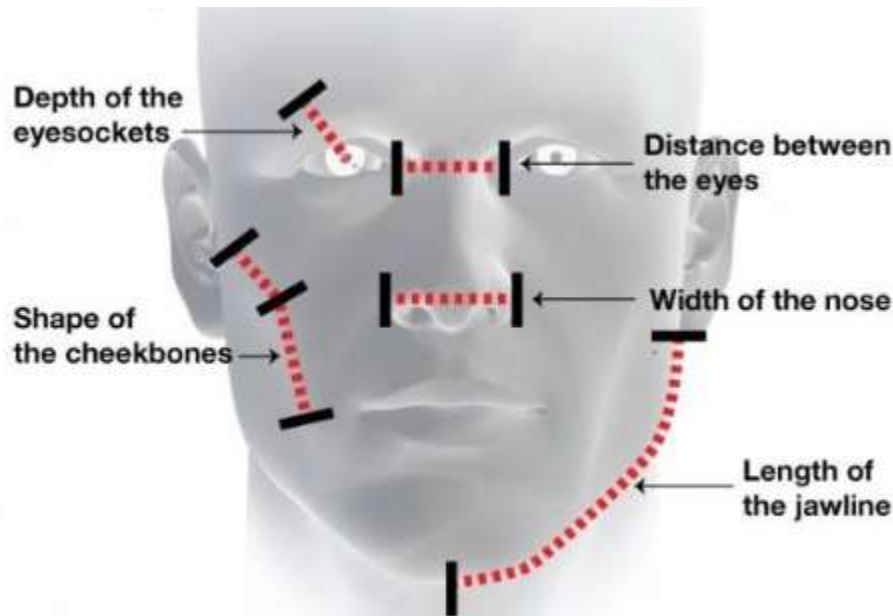


- In the **enrolment phase**, a person's facial image is captured, pre-processed, and analyzed to extract unique features (such as distances between eyes, nose shape, and mouth structure). These features are stored in the system database as a template.
- In the **verification phase**, a new facial image of the user is captured and processed in the same way. The extracted features are then compared with the stored templates in the database. Based on the matching score, the system makes a decision to accept or reject the user.

This process ensures that the system can accurately distinguish between individuals by relying on their unique facial characteristics.

NODAL POINTS IN FACE RECOGNITION

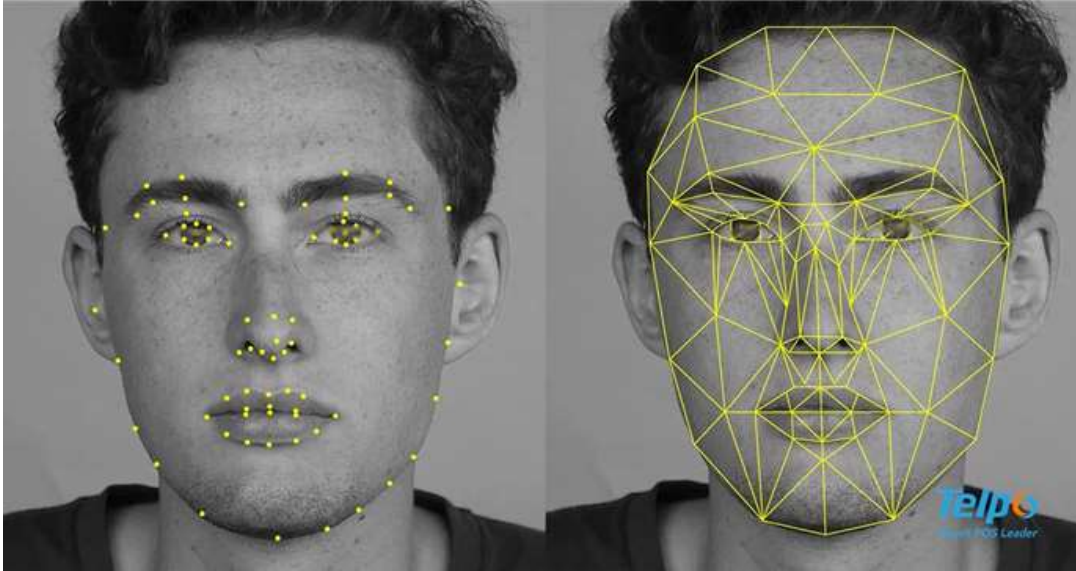
In **facial recognition**, the system relies on **nodal points** (also called *landmarks* or *feature points*) on the human face. These are specific, measurable points on the face that remain relatively constant and are used to create a **faceprint** (a unique mathematical representation of a person's face).



Once the presence of a face is noted, the system can go on to classify, verify, and grant user access. While performing the facial hypothesis, the system analyzes the following features:

- Distance between the eyes
- Distance from the forehead to the chin
- Distance between the nose and mouth
- Depth of the eye sockets
- The shape of the cheekbones
- The contour of the lips, ears, and chin
- Length of the jawline

In facial recognition, nodal points are specific coordinates marked on a face. One point highlights the width of the nose; another can highlight the depth of the eye sockets, and so on. 80 nodal points are analyzed in this way.



Creation of a Faceprint

Categories of Face Recognition methods

Most facial recognition algorithms detect and recognize faces using one of these four methods:

1. Geometric Based

In geometric-based methods, the system distills the face into minuscule geometric data, such as polygons, that represent a person's facial characteristics.

For example, the system analyzes the face. It extracts the positions of distinct features, such as a person's eyes, nose, mouth, and ears, as well as their geometrical data relative to each other. You can also call it a 'feature-based' method.

2. Piecemeal / Holistic

Face recognition systems that use these methods detect and analyze facial data autonomously and independently of each other.

That way, if a person's face is only half visible or is wearing a mask, there is a chance for the system to detect and recognize the face. This is possible when the system can detect the most relevant or distinctive feature of the face (such as a unique eye color, a scar, the distance between the eyes, or the iris pattern) is visible.

3. Appearance-Based / Model-Based

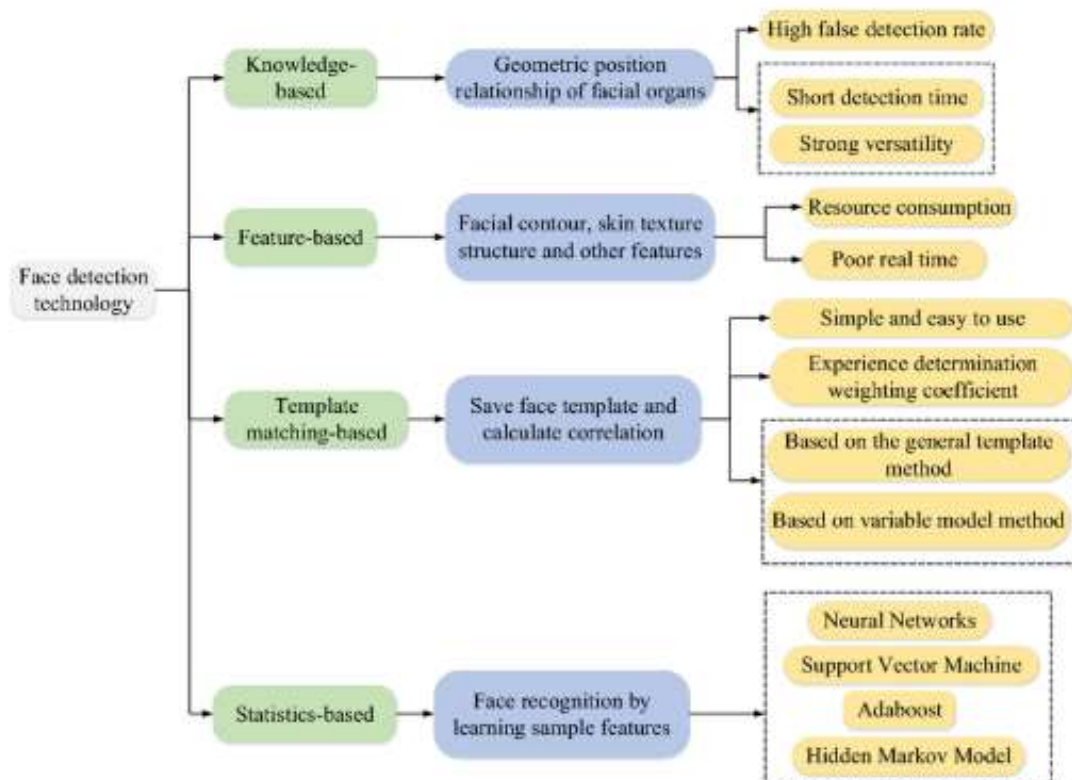
When a system uses these face recognition methods, it considers the image a high-dimensional vector and extracts the pixel density and each pixel's value from the image.

Systems that deal with 2D images often use appearance or model-based face recognition methods, since that's the easiest way to record data.

4. Statistical / Neural Networks Based

In this approach, the system expresses features as patterns. Various statistical tools for extraction and analysis exist, such as Principal Component Analysis (PCA), Discrete Cosine Transform (DCT), and linear discriminant analysis (LDA). An effective system can choose the optimal statistical tool to extract the facial data as effectively as possible.

Neural network-based systems also apply the same process of using statistical tools to extract maximum data for recognition. However, systems using these methods typically deploy AI to find multiple suitable tools and try to combine them to form a hybrid tool for optimal data extraction.



Algorithms for Face Recognition

There are several algorithms that face recognition systems use. Here are 14 of the best ones.

1. Convolutional Neural Network (CNN)

CNN is one of the most popular algorithms in deep learning. It's a type of machine learning algorithm that allows the models to perform classification tasks directly on an image.

CNN has more than a thousand convolutional and pooling layers. Each of these layers learns to detect and process imaging features.

2. Eigenfaces

Eigenfaces is another popular algorithm that determines **face variance** in image data sets. The statistical analysis of a great quantity of face images comes together to make a set of eigenfaces.

This method assigns a mathematical value to a person's facial features. Hence, systems that use this method detect and recognize faces as statistical data where each face has a different percentage of such mathematical values.

3. Fisherfaces

One of the key points that make Fisherfaces a popular and successful alternative to Eigenfaces is its ability to recognize faces from images of varying lighting and facial expressions.

Many consider Fisherfaces to be an extension of Eigenfaces that also takes class labels of the face images into account.

4. Deepface

Deepface uses a CNN model to extract features from the face image. As a result, it is one of the first algorithms to achieve human-level accuracy in an evaluation dataset.

5. Principal Component Analysis (PCA)

The PCA algorithm tries to reduce the data size while also keeping the relevant information intact. The PCA algorithm generates a set of weighted eigenvectors. The

amalgamation of these vectors creates eigenfaces that help systems recognize faces with high accuracy.

6. Haar Cascades

Haar Cascade is an object detection method. This algorithm trains by learning to detect an object of interest in different images and settings.

Once the algorithm is able to successfully detect the object, such as a facial feature, it can look for that feature in different faces to better detect and separate one face from another.

7. Three-Dimensional Recognition

Here, the system uses deep learning to detect and analyze different parameters of a human skull. The skull of each person is slightly different from each other in size and shape. This algorithm allows systems to extract data from the skull's dimensions and use it to tell apart different faces.

This algorithm allows systems to go beyond the two-dimensional facial characteristics and data such as skin color, texture, and the size & shape of the nose and ears. It goes beyond the superficial data of the person's skin and maps the proportions of the skull in 3 dimensions, hence, the name. This allows the system to recognize faces despite the same person wearing glasses, having different make-up, or having facial hair.

8. Skin Texture Analysis

A skin texture analysis algorithm is exactly what the name suggests. It enables face recognition systems to study and detect unique skin parameters like skin color, moles, freckles, or scars to help recognize a face and match it with the database.

On one hand, it requires high-resolution images for the best performance. On the other hand, it can detect faces even if the person has a different hairstyle or facial hair, or is wearing a cap or sunglasses.

9. ANFIS

ANFIS is a type of artificial neural network, and it stands for adaptive neuro-fuzzy interference system. Fuzzy logic means the value of data can be between true (1) and False (0), which is handy in terms of accuracy.

ANFIS combines the advantages of a neural network and fuzzy logic principles to provide an optimal solution. Systems that use ANFIS save time during verification as they are able to classify face image features in the preprocessing stage itself.

10. Local Binary Patterns Histograms (LBPH)

Local Binary Patterns is a simple yet effective way to recognize faces. It marks each pixel by setting the values of its neighboring pixels as thresholds. It stores the result as binary data for future purposes.

This way, the facial recognition system can easily detect and recognize not only faces but other objects through each pixel's relative data with the surrounding ones. The algorithm creates a histogram for each image. As a result, during the verification process, all it has to do is compare the histogram of the input image with the ones in the database.

11. FaceNet

In 2015, researchers from Google developed a face recognition system called FaceNet. They based it on face recognition benchmark datasets and open source pre-trained models are available for third-party implementations.

FaceNet learns to directly map an image and provide compact Euclidean data as a result. This allows the system to store the data as distances that directly correspond to a measure of face similarity.

12. Support Vector Machine (SVM)

SVM is a machine-learning algorithm that is effective at distinguishing faces in images. SVM is a kernel method that excels in a variety of tasks such as text and image classification, handwriting identification, face identification, and anomaly detection.

Classification is the main feature of an SVM algorithm, and it recognizes faces effectively by applying linear and nonlinear training models for face recognition.

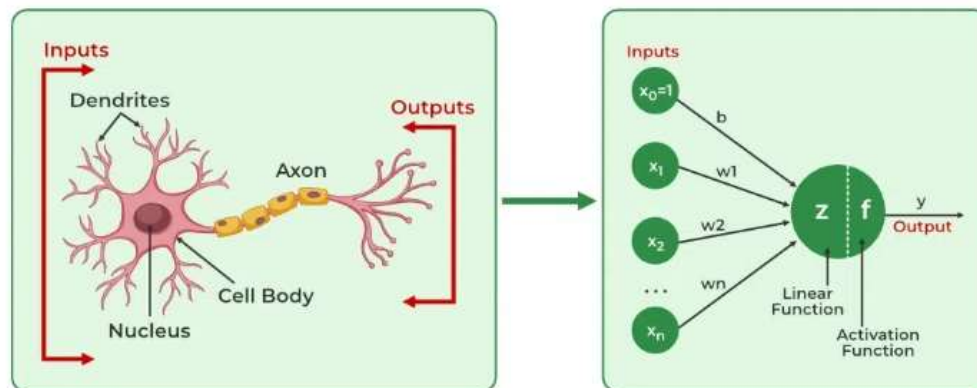
These are the various algorithms you can use for your business depending on their use cases, effectiveness, and overall performance. Speaking of, you can compare the performance of such algorithms through an evaluation dataset.

Neural Network

Neural networks are machine learning models that mimic the complex functions of the human brain. These models consist of interconnected nodes or neurons that process data, learn patterns and enable tasks such as pattern recognition and decision-making.

Neural networks are capable of learning and identifying patterns directly from data without pre-defined rules. These networks are built from several key components:

1. **Neurons:** The basic units that receive inputs, each neuron is governed by a threshold and an activation function.
2. **Connections:** Links between neurons that carry information, regulated by weights and biases.
3. **Weights and Biases:** These parameters determine the strength and influence of connections.
4. **Propagation Functions:** Mechanisms that help process and transfer data across layers of neurons.
5. **Learning Rule:** The method that adjusts weights and biases over time to improve accuracy.

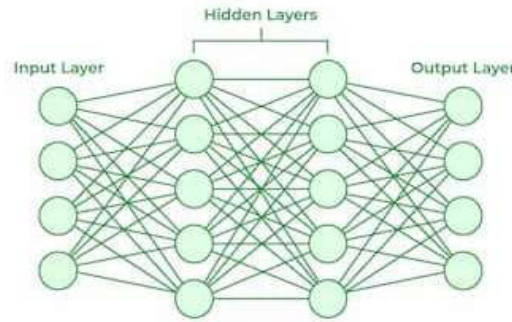


Layers in Neural Network Architecture

1. **Input Layer:** This is where the network receives its input data. Each input neuron in the layer corresponds to a feature in the input data.
2. **Hidden Layers:** These layers perform most of the computational heavy lifting. A neural network can have one or multiple hidden layers. Each layer

consists of units (neurons) that transform the inputs into something that the output layer can use.

3. **Output Layer:** The final layer produces the output of the model. The format of these outputs varies depending on the specific task like classification, regression.



- A neuron computes:

$$y = f\left(\sum_{i=1}^n w_i x_i + b\right)$$

where:

- x_i = input features
- w_i = weights
- b = bias
- $f(\cdot)$ = activation function

Without bias the equation becomes

$$y = f\left(\sum_{i=1}^n w_i x_i\right)$$

Role of Neural Network in Face Recognition Systems

Neural networks (especially **deep learning models**) play a **central role** in modern face recognition. They automate the process of extracting features, learning patterns, and matching faces much more effectively than traditional methods.

1. Feature Extraction

- Earlier methods (like PCA or LDA) required hand-crafted features (e.g., eigenfaces).
- Neural networks, especially Convolutional Neural Networks (CNNs), automatically learn important features such as:
 - Edges, textures, and contours (low-level features)
 - Eyes, nose, mouth, facial geometry (mid-level features)
 - Unique identity features (high-level abstract representations)

This hierarchy makes the representation very robust to lighting, pose, and noise.

2. Dimensionality Reduction & Representation

- Neural networks transform a face image into a compact vector (embedding) in a high-dimensional space.
- Example: A 128-dimensional embedding from FaceNet or a 512-dimensional one from ArcFace.
- Faces of the same person end up close together in this space, while faces of different people are far apart.

3. Face Matching / Verification

- Neural networks help compare two face embeddings using a distance metric (e.g., Euclidean or cosine similarity).
- If the distance is below a threshold it is taken as same person.
- If above it is considered as different persons.

4. Robustness to Variations

Neural networks are trained on massive datasets with faces under different:

- Lighting conditions
- Facial expressions
- Aging effects
- Camera angles / poses

This makes recognition highly accurate in real-world applications.

5. End-to-End Learning

- With neural networks, the entire pipeline (from raw image → feature extraction → classification/matching) can be learned end-to-end.
- Some of the modern models are DeepFace (Facebook), FaceNet (Google), VGGFace, ArcFace.

Advantages of Facial Recognition

Facial recognition has numerous advantages, with the five most critical being:

1. High Accuracy:

The recognition accuracy of next-generation facial recognition engines has surpassed that of human eyes.

2. Fast Recognition Speed:

It can quickly complete facial identity verification within 0.5 seconds, extremely convenient.

3. Contactless:

Compared to other authentication methods (fingerprint scanning), facial recognition is more convenient and hygienic due to its zero-contact nature.

4. Enhanced Information Security:

By integrating with network surveillance cameras and access control systems, it can strictly regulate personnel access, protecting company assets and personnel safety. It also strengthens system login procedures, preventing attacks from hackers.

5. Wide Range of Applications:

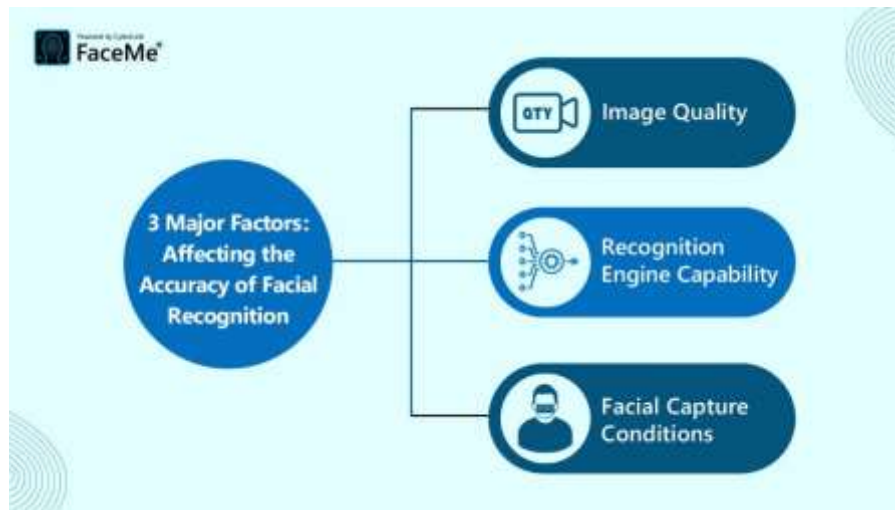
Facial recognition is flexible and can be applied to many applications, such as security, finance, retail, offices, healthcare, and public sectors.

Accuracy of Facial Recognition

The accuracy of facial recognition can be assessed by a low false non-match rate (FNMR) and an extremely low false match rate (FMR).

The false match rate (FMR) indicates the rate at which a person is incorrectly identified as someone else.

The false non-match rate (FNMR) represents the failure to match two images belonging to the same person.



3 Factors Affecting Facial Recognition Accuracy

1. Recognition Engine Capability:

The performance of a recognition engine depends on the technology of the vendor that developed it. This includes the design of the AI model architecture, the volume of training data, the diversity of training data (adequately covering various races, genders, ages, facial angles, lighting conditions, and facial obstructions), and the tuning of parameters during the training process.

2. Image Quality:

Image quality encompasses camera resolution (e.g., 720p), shutter speed, focusing ability, and image quality (e.g., noise and artifacts).

3. Facial Capture Conditions:

This includes the size of the face, lighting conditions on the face (e.g., no backlighting or overexposure), the angle of the face (e.g., a frontal view with rotation less than 50 degrees), and whether the face is partially obstructed (e.g., wearing a mask or sunglasses).

Among these factors, the capability of the facial recognition engine is the most critical. In order to evaluate the performance of different recognition engines, rigorous testing methods and large datasets are needed. For example, the National

Institute of Standards and Technology (NIST) conducts Facial Recognition Technology Evaluations (FRTE) using standardized metrics, execution environments, and API specifications to compare various facial recognition algorithms. The NIST FRTE dataset includes a wide variety of facial image types, with volumes reaching millions in each category.